

Vignesh Vaikundaraman

Research Interests

Star and Planet Formation, Circumstellar Disks, Dust growth and dynamics, Astrochemistry, Astrophysical Fluid Dynamics, Computational Astrophysics

Education

- Feb 2023 – Nov 2026 (Expected) **PhD Student**, *Max Planck Institute for Solar System Research, Göttingen, Germany*,
Supervisor: Dr Joanna Drażkowska,
Thesis: **Dust growth in circumstellar disks using Monte Carlo methods**
Member of the International Max Planck Research School (IMPRS) for Solar System Sciences at the University of Göttingen
- Oct 2019 – May 2022 **M.Sc Physics**, *Ludwig-Maximilians-Universität München, Germany*,
Thesis Supervisor(s): Dr Til Birnstiel, Dr Joanna Drażkowska,
Thesis: Destruction of refractory carbon in protoplanetary disks using Monte Carlo simulations
- July 2015 – June 2019 **B.Tech Mechanical Engineering**, *SASTRA Deemed University, India*,
Thesis: Topology and Stochasticity of Magnetic Fields: A numerical investigation,
Desh-Videsh Scholarship to pursue Bachelor's Thesis in Johns Hopkins University

Research Visits

- Feb 2019 – June 2019 **Research Student**, *Johns Hopkins University, Baltimore, USA*,
Supervisor: Dr Ethan Vishniac
- Nov.2018 – Jan 2019 **Research Intern**, *Indian Institute of Science Education and Research(IISER) , Tirupati, India*,
Supervisor: Dr Jessy Jose

Technical Skills

- Astrophysical Codes **mcdust**, **dustpy**, **FARGO3D**, **RADMC3D**,
Co-developer and maintainer of the open-source code [mcdust](#),
Gained experience in post-processing hydrodynamical codes with dust growth simulations,
Basic experience in post processing hydrodynamical/dust codes with radiative transfer codes.
Experience in benchmarking different astrophysical codes to validate the different aspects of the codes.
- Programming Languages **FORTRAN**, **Python**, **MATLAB**, **Bash**,
Code development in **FORTRAN** and analysis in **Python**
- HPC Tools **SLURM**, **OpenMP**, **MPI**, **profiling tools**,
Gained experience in optimising codes for HPC systems and benchmarking code across different HPC systems.

Supervision Experience

- May – Jul 2025 **Co-supervisor**, *Emilio Saldaña Markus*, Summer Intern,
Topic: Connecting hydro code **FARGO3D** with dust growth code **mcdust**
- May – July 2023 **Co-supervisor**, *Jayashree Narayan*, Summer Intern,
Topic: Formation of Close-in Exoplanets by Pebble Accretion

Teaching Experience

- WS 2024/25 **Introduction to Astrophysics, University of Göttingen,**
Taught the course in English in parallel to the main German course, designed and adapted course materials to suit the students who cannot speak German
- SS 2024 **Planetary Systems, Max Planck Institute for Solar System Research,**
Tutor for the course exercises
- WS 2023/24 **Introduction to Astrophysics, University of Göttingen,**
Tutor for the course exercises

Talks and Conferences

Invited Talks

- Nov 2025 GLOBE Institute, Copenhagen, Denmark
- Nov 2025 Tata Institute for Fundamental Research, Mumbai, India
- Nov 2025 Institute for Radio Astronomy and Astrophysics (IRyA), UNAM, Mexico (Online)
- Feb 2025 National Institute of Science Education and Research, Bhubaneswar, India

Conferences

- May 2026 MIAPbP program: Planet Formation across various Environments, and Epochs, Garching, Germany (talk)
- June 2025 Origins of Solar Systems, Mt Holyoke College, USA (Poster)
- Feb 2025 Astronomical Society of India Annual Meeting 2025, Bhubaneswar, India (Talk)
- Dec 2024 The Minor Bodies of the Solar System: Space Missions, Observations and Theory, Ringberg, Germany (Talk)
- Nov 2024 Exoplanets - Diversity, Formation and Evolution 2024, Göttingen, Germany (Talk)
- Sept 2024 EXOHOST Summer School, Tartu, Estonia
- July 2024 EAS Annual Meeting, Padova, Italy (Talk)
- August 2023 SPP 1992 Planet formation Summer School, Heidelberg, Germany (Poster)
- April 2023 Protostars and Planets VII Kyoto, Japan (Poster)

Organisational Experience

- Aug 2025 - May 2026 **Member of committee to extend computational infrastructure,**
Contributed to the evaluation and benchmarking of different HPC systems to choose the best combination of systems suitable for the current computational needs, 840,000 EUR awarded to the institute by the Max Planck Society to expand its computational infrastructure
- Jan 2024 - Jan 2026 **Co-ordinator of the Max Planck PhDnet Open Science Working Group,**
Co-organiser of the Max Planck Open Science Ambassadors Meeting, Part of the working group to make sure Early Career researchers are represented in the Max Planck Open Science policies

Languages

- English *Fluent*
- Tamil, Telugu *Mother Tongue*
- German *A2*

Outreach

Member of [The EqualSTEM project](#): Mentoring minoritised students to pursue higher education in STEM

[Science Communication in Tamil with Phyfron](#): A virtual seminar on planet formation in Tamil

References

Dr. Joanna Drazkowska, *Research Group Leader*,
Max Planck Institute for Solar System Research,
Email: drzkowska@mps.mpg.de

Prof. Dr. Til Birnstiel, *Professor*,
Ludwig-Maximilians-Universität München,
Email: til.birnstiel@lmu.de